

PAPER_201: UQFF Gravitational Waves — Chirp Mass, QNM, BZ, Orbital Decay, and Kilonova

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$$F_U(r, t) = \sum_{i=1}^4 U_{gi} + U_m + U_A - U_{bi}, \quad \kappa = 5.0 \times 10^{-4} \text{ day}^{-1}, \quad [SSq] = 0.57$$

$$h_{\text{UQFF}}(t) = h_{\text{GR}}(t) \cdot (1 - U_{bi}/F_U) \cdot e^{-\kappa t}, \quad \kappa = 5.0 \times 10^{-4} \text{ day}^{-1}$$

Abstract

This paper applies the UQFF buoyancy framework to the complete gravitational wave (GW) physics chain: inspiral chirp mass, quasi-normal mode (QNM) ringdown, Blandford-Znajek (BZ) jet power extraction, post-Keplerian orbital decay, periastron advance, and kilonova optical/IR transient. The F_UBii/Um framework unifies these into a single UQFF operator chain: inspiral ? plunge ? ringdown ? jet ? remnant (kilonova). Numerical coefficients from LIGO GWTC-4.0 and EHT observations calibrate the constants.

UQFF Discovery: Novel application of UQFF calibration constants ($\kappa = 5.0 \times 10^{-4} \text{ day}^{-1}$, $[SSq] = 0.57$) uniquely enabling this analysis — establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. GW inspiral: Chirp Mass

$$F_{\text{UBii, chirp}} = F_{\text{rel}} \times (?? / E_{\text{LEP}}) \times Q_{\text{wave}} \times [dE/dt = -(32/5)G4\mu^2M^3/(c5r5)]$$

$$?? = (m1m2)^{3/5}/(m1+m2)^{1/5} \quad (\text{chirp mass, solar masses})$$

$$?? = (c^3/G) \cdot (5/96 \cdot p^{-8/3} \cdot f^{-11/3} \cdot ?)^{3/5} \quad (\text{from observed GW frequency drift})$$

$$U_{m, \text{chirp}}(f) = \mu(?_{\text{vac}}) \cdot (1 - e^{-?t}) \cdot (32/5 \cdot G4 \cdot \mu^2 M^3 / c5r5)$$

$$\text{Calibration: GW150914 } ? ?? \sim 28.3 M_{\text{?}}; \text{ GW170817 } ? ?? \sim 1.188 M_{\text{?}}$$

2. QNM Quasi-Normal Mode Ringdown

$$f_{\text{QNM}} = c^3/(2pGM_f) \cdot f(a_f)$$

where the Berti et al. fitting function:

$$f_{\text{QNM}} = (c^3/2pGM_f) \cdot [0.3737 + 0.088 \cdot a_f + \dots] \quad (l=2, m=2 \text{ dominant mode})$$

Decay time:

$$t_{\text{QNM}} = GM_f/c^3 \cdot g(a_f) \quad (g \sim 2-10 \text{ M for } a_f \sim 0.69)$$

$$F_{UBii,qnm} = -F_{rel} \times (f_{QNM} / E_{LEP}) \times Q_{wave} \times e^{-t/t_{QNM}}$$

$$Um,qnm(a) = \mu(?_{vac}) \cdot a \cdot c^3 / (2GM) \times (1 - e^{-t}) \cdot [l=2 \ s=-2 \text{ perturbation}]$$

$$Um,damp(a) = \mu(?_{vac}) \cdot Q_{factor} \times (1 - e^{-t}) \cdot [Q \sim 10 \text{ for dominant mode}]$$

Calibration: GW150914 final BH $M_f \sim 62 M_\odot$, $a_f \sim 0.67$? $f_{QNM} \sim 251$ Hz

3. Blandford-Znajek Jet Power

$$P_{BZ} = (1/32) \cdot B^2 \cdot R_H^4 \cdot \Omega_H^2 / c \quad (\text{BZ original form})$$

Updated EHT form:

$$P_{BZ,EHT} = (??/16p) \cdot F_{BH}^2 \cdot \Omega_{BH}^2 / c$$

where:

$$? \sim 0.044 \quad (\text{numerical factor from GRMHD})$$

$$F_{BH} = B \cdot p \cdot r_H^2 \quad (\text{BH magnetic flux, EHT M87* calibrated})$$

$$\Omega_{BH} = a \cdot c^2 / (2r_H \cdot c) \quad (\text{angular velocity})$$

$$\text{For M87*}: B \sim 1\text{--}30 \text{ G} \quad ? \quad P_{BZ} \sim 10^4 ?^3 \text{ erg/s}$$

$$F_{UBii,bz} = F_{rel} \times ((1/32) \cdot B^2 \cdot R_H^4 \cdot \Omega_H^2 / c / E_{LEP}) \times Q_{wave}$$

$$F_{UBii,bz2} = F_{rel} \times ((??/16p) \cdot F_{BH}^2 \cdot \Omega_{BH}^2 / c / E_{LEP}) \times Q_{wave}$$

$$Um,bz(a) = \mu(?_{vac}) \cdot \text{Power} \quad ? \quad B^2 \Omega_H^2 \cdot R_H^4 \times (1 - e^{-t})$$

$$Um,bz2(a) = \mu(?_{vac}) \cdot (??/16p) F_{BH}^2 \cdot \Omega_{BH}^2 / c \times (1 - e^{-t})$$

4. Post-Keplerian Orbital Decay

GW orbital decay rate (Peters formula, GR 2.5PN):

$$?_b = -(192p/5) \cdot (P_b/2p)^{-5/3} \cdot (G^{??\{5/3\}/c^3})^{5/3} / (P_b)^{5/3} \cdot f(e)$$

$$f(e) = [1 + (73/24)e^2 + (37/96)e^4] \cdot (1 - e^2)^{-7/2}$$

$$F_{UBii,orbdec} = -F_{rel} \times (?_b/P_b = dE/dt / E_{LEP}) \times Q_{wave}$$

$$Um,orbdec(e) = \mu(?_{vac}) \cdot [-dE_{GW}/dt] \times (1 - e^{-t})$$

Calibration: Hulse-Taylor PSR B1913+16

$$?_b,obs \sim -2.422 \times 10^{-12} \quad (\text{dimensionless})$$

$$?_b,GR \text{ prediction fits to } < 0.1\%$$

5. Post-Keplerian Periastron Advance

$$?? = 3 \cdot (P_b/2p)^{-5/3} \cdot (G(m_1+m_2)/c^3)^{2/3} / (1 - e^2)$$

$$F_{UBii,peri} = F_{rel} \times (?? / E_{LEP}) \times Q_{wave} \times (G(m_1+m_2))^{2/3} \cdot (1 - e^2)^{-1}$$

$$Um,peri(a) = \mu(?_{vac}) \cdot (\text{Kepler: } a^3/P^2 = GM/(4p^2)) \times (1 - e^{-t})$$

Calibration: PSR B1913+16 $\dot{\Omega} = 4.226^\circ/\text{yr}$ (measured) vs $4.226^\circ/\text{yr}$ (GR)

6. Kilonova Optical/IR Transient

$$L_{\text{peak}} \sim 104^1 \cdot (M_{\text{ej}}/0.01 M_\odot) \cdot (v_{\text{ej}}/0.1c) \cdot (\kappa/1 \text{ cm}^2/\text{g})^{-1} \text{ erg/s}$$

$$\text{Peak timescale: } t_{\text{peak}} \sim v(3M_{\text{ej}}/(4\pi c v_{\text{ej}}))$$

$$F_{\text{UBii,kilo}} = F_{\text{rel}} \times (M_{\text{ej}} \cdot v_{\text{ej}} \cdot c / (\kappa \cdot L_{\text{peak}}) / E_{\text{LEP}}) \times Q_{\text{wave}}$$

$$U_{\text{m,kilo}}(t) = \mu(\kappa_{\text{vac}}) \cdot (\text{Diffusion } t_d^2 = 3M/(4\pi c v_{\text{ej}}^2)) \times (1 - e^{-t/t_d})$$

Calibration: AT2017gfo (GW170817 neutron star merger):

$$M_{\text{ej}} \sim 0.05 M_\odot, v_{\text{ej}} \sim 0.15c, \kappa \sim 1-5 \text{ cm}^2/\text{g}$$

$$L_{\text{peak}} \sim \text{few} \times 104^1 \text{ erg/s (r-process nucleosynthesis powered)}$$

7. UQFF GW Chain Unification

The entire GW compact binary lifecycle maps onto UQFF operators:

1. Inspiral $\dot{F}_{\text{UBii,chirp}} + U_{\text{m,chirp}}$ [orbital energy loss, GW frequency sweep]
 $\dot{\Omega}$ plunge/merger
2. Ringdown $\dot{F}_{\text{UBii,qnm}} + U_{\text{m,qnm}}$ [ringing BH, quenches exponentially]
 $\dot{\Omega}$ spin-down
3. Jet launch $\dot{F}_{\text{UBii,bz}} + U_{\text{m,bz}}$ [magnetically extracted power]
 $\dot{\Omega}$ r-process
4. Remnant $\dot{F}_{\text{UBii,kilo}} + U_{\text{m,kilo}}$ [neutron star ejecta optical transient]

Long-term:

5. Orbital decay $\dot{F}_{\text{UBii,orbdec}} + U_{\text{m,orbdec}}$ [binary evolving toward merger]
 6. Periastron $\dot{F}_{\text{UBii,peri}} + U_{\text{m,peri}}$ [GR precession measured]
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8. Numerical Summary

Process	UQFF Parameter	Calibration System
Chirp mass $\dot{\Omega}$	$28.3 M_\odot$	GW150914
QNM freq	251 Hz	GW150914
BZ power	$104^2 \text{ to } 104^3 \text{ erg/s}$	M87* EHT
$\dot{\Omega}_{\text{b}}/\dot{\Omega}_{\text{b,GR}}$	<0.1% deviation	PSR B1913+16
$\dot{\Omega}$ match	$4.226^\circ/\text{yr}$	PSR B1913+16
Kilonova L	$\text{few} \times 104^1 \text{ erg/s}$	AT2017gfo

9. References

- grok_share_7514fe.txt lines 6060–6080 (BB_C_Equations items 1292–1300, 1556–1560)

- PAPER_198: F_UBii Taxonomy Part 1 (QNM/BZ/Sedov)
- PAPER_200: Um Universal Magnetism Catalogue
- LIGO GWTC-4.0 (2025 catalog)
- EHT M87* 2019 polarization papers